

FrontierAtlas / GraphOne — Production Intelligence System Architecture

System Design Document: High-Throughput Ingestion (500k Scale), Resilient Multi-Tier LLMs, Anti-Bot Camouflage & Knowledge Graph Persistence

1. Distributed Ingestion Pipeline & Scaling to 500k+ Records

To scale entity extraction from 1,000 to 500,000+ records without pipeline degradation, the system decouples crawler discovery, LLM inference, and graph persistence into an event-driven microservices architecture:

- **Domain-Partitioned Kafka Queues:** Target discovery jobs are partitioned by root domain (e.g., *arxiv.org*, *techcrunch.com*), allowing independent concurrency limits and politeness controls per source without global throughput bottlenecks.
- **Stateless Async Workers:** Python worker nodes running aiohttp and asyncio auto-scale horizontally on Kubernetes (HPA) governed by consumer queue lag metrics.
- **Batch Checkpointing & Memory Safety:** Workers flush records incrementally to durable staging storage every 100 items with non-blocking I/O, preventing memory exhaustion and ensuring zero data loss on unexpected restarts.

2. Anti-Bot Navigation & Stealth Harvesting

- **TLS/JA4 Fingerprint Camouflage:** Employs curl_cffi to match standard Chrome/Safari TLS cipher suites and HTTP/2 settings, preventing automated flags from Cloudflare Turnstile, DataDome, and Akamai.
- **Headless Browser Management:** Aggressive JavaScript pages use Playwright Async wrapped with playwright-stealth, randomized viewport geometries, human mouse bezier paths, and rotating residential IP proxies.
- **Politeness Rate Shaping:** Adaptive token-bucket rate limiters dynamically adjust per-domain request intervals based on upstream HTTP response codes (e.g., automatically backing off upon encountering HTTP 429).

3. Resilient Multi-Tier LLM Orchestration (413 & 429 Mitigation)

- **413 Payload Too Large Prevention:** Raw HTML documents undergo structural DOM pruning (stripping scripts, styles, SVG vectors, and base64 binaries) followed by sentence-aware window chunking to cap prompt payloads under token limits. If an upstream 413 is received, chunks dynamically halve and retry.
- **429 Rate-Limit Exponential Backoff:** Rate limits trigger full exponential backoff with randomized jitter [$T = 2^{\text{attempt}} + \text{Uniform}(0.5, 1.5)$], preventing synchronized thundering-herd retries across concurrent worker nodes.
- **Circuit-Breaker Failover Chain:** Seamlessly cascades across providers: **Tier 1 (Gemini 1.5 Flash)** → **Tier 2 (Groq Llama-3-70B)** → **Tier 3 (DeepSeek-V3)**. If a tier exhausts retries, requests fail over automatically without worker interruption.

4. 24-Hour Freshness Tracking & Deduplication Engine

- **RedisBloom Fast-Path Deduplication:** Distributed Bloom filters check URL and title signatures in sub-millisecond O(1) time (false-positive rate < 0.1%), preventing redundant parsing across worker nodes.
- **Strict UTC Delta Window:** Ingested publication timestamps are parsed across RFC-822, ISO-8601, and natural-language formats, normalizing to UTC. Records where $\Delta t = \text{now_utc} - t_{\text{published}} > 24.0h$ are rejected at the ingestion boundary.
- **Content-Hashing Identity:** Entity records generate SHA-256 signatures over normalized attributes (name + date + url) with a 7-day sliding TTL to prevent duplicate insertions.

5. Primary Storage & Intelligence Graph Architecture

- **PostgreSQL (ACID Primary Store):** Stores schema-validated entities, raw payload backups, ingestion telemetry, and historical GitHub star metrics.
- **Neo4j Property Graph Layer:** Models dynamic industry relations: (Startup)-[:BUILT]->(Product), (Startup)-[:PUBLISHED]->(Paper), (Paper)-[:HAS_REPO]->(GitHubRepo), and (Startup)-[:OPENED]->(Job).
- **Qdrant Vector Database:** Stores 1536-dimensional embeddings for research abstracts and news articles to enable semantic deduplication, topic clustering, and retrieval.